HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented **Python**-based prompt engineering and retrieval augmented generation (RAG) pipelines for an AI financial insights product, processing ~1,500 customer queries daily.
- Built and deployed **FastAPI** inference services on **Azure Container Instances** to serve LLM-generated financial advice, achieving p95 latency below **200ms** for customer-facing interactions.
- Developed **TypeScript** and **Node.js** backend services to manage conversational state and tool use for an AI banking assistant, integrating with a **PostgreSQL** feature store.
- Constructed evaluation frameworks for AI features, comparing LLM outputs against ground truth transaction data and measuring user-defined helpfulness metrics, leading to a **15%** improvement in relevant suggestions.
- Collaborated with AI experience leads to translate product designs into actionable AI feature implementations, iterating on prompt strategies based on observed model behaviour and customer data.
- Implemented production safeguards for AI-driven financial guidance, ensuring answers were grounded in customer data, clear of direct financial advice, and included defined human handover points.
- Containerized AI services using **Docker** and deployed them to an **Azure Kubernetes Service (AKS)** cluster, configuring basic resource requests and limits for predictable performance.

SKILLS

AI/ML – Large Language Models (LLMs), Retrieval Augmented Generation (RAG), Prompt Engineering, Model Evaluation, Generative AI, AI Safeguards, Conversational AI

Languages – Python, TypeScript, SQL, Java

Backend Frameworks – FastAPI, Node.js, NestJS, Spring Boot

Databases – PostgreSQL, Vector Databases (conceptual)

Cloud Platforms – Azure (AKS, Container Instances, Azure Functions), AWS (EC2, S3, Lambda, CloudWatch)

DevOps & Tools – Docker, Kubernetes, Git, Jenkins, Azure DevOps, CI/CD

PROJECTS

AI Financial Insights Engine

- Developed a **Python** RAG pipeline that ingests and indexes bank transaction data into a vector database, enabling contextual retrieval for an AI financial advisor.
- Implemented prompt chaining and tool use capabilities within **LangChain** to allow the AI to query structured data (**SQL**) and generate personalized financial summaries for small business owners.
- Designed and shipped an evaluation suite to assess the accuracy and helpfulness of AI-generated insights, comparing against a baseline of **500** manually generated reports.

Customer Conversational AI Feature

- Built a **TypeScript**-based API service using **NestJS** to manage user sessions and orchestrate calls to an LLM for a customer support chatbot.
- Integrated a fine-tuned language model for specific financial domain queries, achieving a **70%** reduction in irrelevant responses compared to a general-purpose model.
- Deployed the service to **Azure Functions**, handling approximately **2,000** requests per day during peak load.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)